Omid Mokhtari

Ph.D. Student in Electrical Engineering, University of Vermont
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Research Interests

Power system operation and planning; mathematical optimization; graph theory; network reduction; sensor placement.

Education

University of Vermont Burlington, VT Aug. 2023–Present
Ph.D. in Electrical Engineering (expected Fall 2027)
Advisors: Samuel Chevalier and Mads R. Almassalkhi.

Sharif University of Technology Tehran, Iran Sept. 2017–Jan. 2020
M.Sc. in Electrical Engineering – Power Systems
Thesis: "Co-planning of Multiple Merchant Energy Storage Systems and Transmission Systems." Advisor: Seyed Hamid Hosseini.

Imam Khomeini International University of Qazvin Qazvin, Iran Sept. 2012–July 2017
B.Sc. in Electrical Engineering – Power Systems
Final project: "Investigation of Wind Turbines Dynamic Behavior During a Fault, Using DIgSILENT." Advisor: Navid Ghaffarzadeh.

Research Experience

Doctoral Research, University of Vermont Aug. 2023–Present

Distribution-network reduction

- Developed **Opti-KRON**, an optimization-based method for reducing radial distribution feeders while preserving radial topology and limiting voltage-profile errors.
- Extended Opti-KRON to realistic three-phase feeders and developed a GPU-based algorithm to improve scalability.

Network reduction via community detection

- Investigated community-detection-based decomposition for solving larger network-reduction problems.

Transmission-network reduction (ongoing)

- Developing network aggregation methods for DC optimal power flow and planning, with emphasis on reduced-model feasibility and line-flow constraints.

Sensor placement (ongoing)

- Optimizing sensor locations in radial feeders to minimize worst-case voltage-magnitude error under sensor-budget constraints and uncertain line parameters.

Master's Research, Sharif University of Technology 2017–2020

- Modeled competition among merchant energy storage investors through bilevel electricity-market problems and an EPEC formulation; investigated a diagonalization method for equilibrium analysis.

Publications

Peer-reviewed journal articles

[**J1**] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Structure-preserving optimal Kron-based reduction of radial distribution networks," *IEEE Transactions on Power Systems*, early access, 2026. doi:10.1109/TPWRS.2026.3726028.

[**J2**] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Optimal Kron-based reduction of networks (Opti-KRON) for three-phase distribution feeders," *Electric Power Systems Research*, vol. 263, Art. no. 113615, 2027 (PSCC 2026 paper). doi:10.1016/j.epsr.2026.113615.

Preprints

[**P1**] O. Molodchyk, **O. Mokhtari**, S. Chevalier, M. R. Almassalkhi, and T. Faulwasser, "Data-driven power flow for radial distribution networks with sparse real-time data," *arXiv preprint*, arXiv:2604.08403, 2026. [Link]

[**P2**] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Enhancing scalability of optimal Kron-based reduction of networks (Opti-KRON) via decomposition with community detection," *arXiv preprint*, arXiv:2407.02679, 2024. [Link]

Research Software

Opti-KRON: Public research code for optimization-based network reduction. GitHub repository.

Conference and Poster Presentations

2026. "Structure-preserving optimal Kron-based reduction of radial distribution networks". *PowerUp, Boulder, CO (poster).*

2024–2026. Three posters on community detection, network reduction and radialization, and three-phase Opti-KRON. *University of Vermont Student Research Conference.*

2024. "Optimal network reduction". *IEEE Power & Energy Society General Meeting, Seattle, WA (poster).*

Academic Service

Peer reviewer: *IEEE Transactions on Power Systems*, *Electric Power Systems Research*, and American Control Conference (ACC).

Technical Skills

Programming and tools: Julia, Python, MATLAB, GAMS, LaTeX.

Optimization solvers: Gurobi, MOSEK, IPOPT, CPLEX.

Power-system software: DIgSILENT, OpenDSS, PSS/E.